

Abinaya Rengarajan

M.Sc. | Ph.D. (Physics) | Ph.D. (Optoelectronics & Nanostructure Science)

Research Assistant Professor / Post-Doctoral Fellow — Thermoelectrics

Thermoelectric Research Lab, Dept. of Inorganic Chemistry, AGH University of Krakow
al. Mickiewicza 30, 30-059 Kraków, Poland

abynayarangarajan@gmail.com | abinaya@agh.edu.pl | +91 7639172287 / +48 729639500

scholar.google.com/citations?user=NeBT4L0AAAAJ

29 Publications	564 Citations	43 Intl. Conf.	14 h-index
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RESEARCH PROFILE

Experimentalist specializing in synthesis, fabrication, and characterization of low-dimensional **transition metal dichalcogenides (TMDCs)** and chalcogenide materials for **thermoelectric energy harvesting, photodetectors, gas sensors, and solar cells**. Research spans 2D materials (MoS_2 , WS_2 , SnS_2), flexible devices, and inorganic–organic hybrid nanocomposites. Author of **29 peer-reviewed publications** in high-impact journals (Advanced Functional Materials, Journal of Materials Chemistry C, Chemical Engineering Journal, ACS Applied Energy Materials, Applied Physics Letters) with **564 citations** (Google Scholar). Expertise directly aligns with the required fields: *thermoelectrics, solar cells, photodetectors, gas sensors, electrocatalysis, and 2D materials*.

EDUCATION

Ph.D. — Optoelectronics and Nanostructure Science (Graduated) Sep 2022
Shizuoka University, Hamamatsu, Japan | Supervisor: Prof. Masaru Shimomura
Thesis: Evaluation of thermoelectric properties of planar, vertical and radially grown 2D transition metal dichalcogenides

Ph.D. — Physics (Graduated) Nov 2022
SRM Institute of Science and Technology, Chennai, India | Supervisor: Dr. J. Archana
Thesis: Interface-engineering of layered MoS_2 nanostructures for thermoelectric applications

M.Sc. — Physics May 2016
Bharathidasan University, Tiruchirappalli, India | Supervisor: Prof. K. Jeganathan
Thesis: Graphene–Silicon Schottky Junction Solar Cell

PROFESSIONAL EXPERIENCE

Research Assistant Professor / Post-Doctoral Fellow — Thermoelectrics May 2024 – present
AGH University of Krakow (AGH UST), Kraków, Poland (Maternity/parental leave Jan 2025–Jan 2026)
Supervisor: Prof. Dr. Krzysztof Wojciechowski | Thermoelectric Research Laboratory

Post-Doctoral Fellow — Thermoelectrics, Optoelectronics & Device Applications Nov 2021 – Mar 2024
SRM Institute of Science and Technology, Chennai, India
Supervisor: Prof. S. Ponnusamy

HONORS AND AWARDS

- Best Oral Presentation Award (1st) as corresponding author— ICEMSST 2025, VIT Chennai, India.
- Best oral presentation, AFMD 2024, SRM IST, India.
- Best poster presentation (1st), DPRC 2022, SRM IST, India.
- Visiting Research Student - 2022, Cavendish Laboratory, University of Cambridge, UK
- Best poster presentation, DPRC 2021, SRM IST, India.
- Dean's Award of GSST for Scholarly Research Work - 2019, Shizuoka University, Japan
- BANAJ Indian Scholarship (for Ph. D), (April 2019 – March 2021), Japan
- Best Oral Presentation Award (1st), ICONN 2019, SRM IST, India
- Best Poster Presentation Award (1st), ICONN 2019, SRM IST, India

- Best Oral Presentation Award (3rd), NCHSM-2019, SRM IST, India
- Young Researcher Award, ICNERE 2018, Shizuoka University, Japan
- Best Oral Presentation Award, ICNERE 2018, Shizuoka University, Japan,
- Visiting Research student (Oct 2017 to Marc 2018), Shizuoka University, Japan,
- Best Poster Presentation Award (1st), AOSS 2017, SRM IST, India
- Best Poster Presentation Award (1st), National Symposium 2017, Periyar University, India
- University Research Fellow, (2016-2022), SRM IST, India
- Winter Internship program at IIA Kodaikanal Solar Observatory, (18- 28 May), 2015, India.
- SRC Management Prize — 1st place in Mathematics (B.Sc.), 2013, India
- Indian Association of Physics Teachers (IAPT 2012) — 1st place, Tiruchirappalli, India

KEY RESEARCH ACCOMPLISHMENTS

- Achieved record thermoelectric figure-of-merit $zT = 1.18$ in MoS_2 by engineering sulphur vacancies via MoO_3 hole injection — among the highest reported for 2D TMDs (Chemical Engineering Journal, 2021).
- Demonstrated ambipolar Seebeck coefficient in $\text{MoS}_2/\text{MoS}_2\text{-MoO}_2$ graded composites, enabling reversible p/n switching by junction reciprocity (The Journal of Physical Chemistry Letters, 2024).
- Decoupled Seebeck coefficient and electrical conductivity trade-off in few-layer MoS_2 via controlled out-plane lattice dynamics (Applied Physics Letters, 2022).
- Developed flexible thermoelectric nanocomposite based on PEDOT:PSS / $\text{Cu}_{12}\text{XSb}_4\text{S}_{13}$ / MWCNTs achieving $>2\times$ power factor improvement over pristine PEDOT:PSS (AGH, 2025).
- Modulated Fermi energy in few-layer MoS_2 via metal passivation, achieving enhanced near-IR detectivity for photodetectors (Journal of Materials Chemistry C, 2024).
- Demonstrated room-temperature NO_2 gas sensing (ppb-level) using vertically grown SnS_2 nanosheets and SnS_2/SnS 2D heterostructures (Applied Surface Science & Journal of Alloys and Compounds, 2024).
- Constructed a bespoke high-temperature Seebeck coefficient (2-probe) and electrical resistivity (4-probe linear) measurement system calibrated to $>99\%$ accuracy with constantan standard.
- First report on preparation and optimization of highly stable densified $\text{MoS}_2/\text{polyaniline}$ composites for temperature dependent transport property measurements.
- Decoupling trade-off thermoelectric relation in $\text{MoS}_2/\text{polyaniline}$ nanocomposites via optimizing potential barrier height <0.1 eV.
- Track record on optimization of active sites in nanostructures towards high performance photoelectrode, gas sensing, photocatalytic applications via defects/surface/nanoarchitectures-engineering
- Preparation of ultrathin layered nanosheets of MoS_2 with optimized thickness of 15-20 nm for high performance photocatalytic applications

PEER-REVIEWED JOURNAL PUBLICATIONS

562	525	14	18
Total Citations	Since 2021	h-index	i10-index

Note: Author appears as 'R. Abinaya', 'A. Rengarajan', or 'Abinaya Rengarajan' across publications; all refer to the same author. Name in bold. Publications in bold are relevant to the requirement *Author's name in bold where lead/co-lead; ‡ corresponding author.* | [Google Scholar](#)

A. PEER-REVIEWED JOURNAL ARTICLES

2026

1. SS John, MJK Peer, **Abinaya Rengarajan**, A Jayaram, M Navaneethan "Interface and dimensionality driven engineering of carrier transport and phonon dynamics in $\text{MoS}_2/\text{MWCNT}$ nanocomposites." Carbon Letters 36, 753–765 (2026). <https://doi.org/10.1007/s42823-025-01014-1>
2. A Lis, A Sharma, K Zazakowny, **Abinaya Rengarajan**, T Mori, KT Wojciechowski "Free-standing thermoelectric nanocomposite based on PEDOT:PSS, $\text{Cu}_{12}\text{XSb}_4\text{S}_{13}$, and MWCNTs." Journal of Materials Chemistry C 14, 8152–8167 (2026). <https://doi.org/10.1039/D5TC04463D>

2025

3. SS John, **A Rengarajan**, K Chinnadurai, MM Chehimi, SK Eswaran, A Jayaram, N Mani "**Band structure tuning and thermal conductivity suppression via isovalent Te substitution in MoS_2 for thermoelectric applications.**" Journal of Alloys and Compounds 184327 (2025). <https://doi.org/10.1016/j.jallcom.2025.184327>

4. R Abimaheshwari, **R Abinaya**, C Suresh Prasanna, M Navaneethan, S Harish "**Dynamic growth configuration of 2D-SnS₂ nanoflakes for humidity-resistant NO₂ monitoring system: mechanistic insights and environmental applications.**" *Small* 21 (33), 2504996 (2025). <https://doi.org/10.1002/sml.202504996>
5. R Santhosh, **R Abinaya**, S Harish, J Archana, S Ponnusamy, H Ikeda, M Navaneethan "Interface-driven energy filtering effect and phonon scattering in n-type Ag₂Se/polyaniline nanocomposites for enhanced thermoelectric performance." *Carbon Letters* (2025). <https://doi.org/10.1007/s42823-025-00894-6>
6. R Abimaheshwari, T Murugadass, **R Abinaya**, M Navaneethan, S Harish "**Fabrication of SnS₂/Si heterostructure for ultra-high selective and rapid room temperature NO₂ gas detection with enhanced carrier mobility.**" *Sensors and Actuators B: Chemical* 428, 137165 (2025). <https://doi.org/10.1016/j.snb.2025.137165>
7. C Archana, **R Abinaya**, N Mani, HS Krishnan "**Strain effect in the layered MoS₂-rGO heterostructure with enhanced performance for flexible thermoelectric applications.**" *ACS Applied Energy Materials* 8 (3), 1589–1597 (2025). <https://doi.org/10.1021/acsaem.4c02700>

2024

8. R Abimaheshwari, **R Abinaya**, M Navaneethan, S Harish "Enhanced sensitivity and selectivity of ZnSe/PANI nanocomposite for low-ppm NO₂ detection at room temperature gas sensor application." *Surfaces and Interfaces* 55, 105295 (2024). <https://doi.org/10.1016/j.surfin.2024.105295>
9. **A Rengarajan**, MJK Peer, H Santhana Krishnan, A Jayaram, N Mani "**Ambipolar Seebeck coefficient in MoS₂/MoS₂-MoO₃ graded composites by junction reciprocity.**" *The Journal of Physical Chemistry Letters* 15 (49), 12060–12067 (2024). <https://doi.org/10.1021/acs.jpclett.4c02683>
10. R Abimaheshwari, **R Abinaya**, J Archana, C Muthamizhchelvan, M Navaneethan, S Harish "**In-situ growth of vertically stacked 2D SnS₂/SnS heterostructure for highly sensitive and rapid room temperature NO₂ gas sensor.**" *Journal of Alloys and Compounds* 1001, 175002 (2024). <https://doi.org/10.1016/j.jallcom.2024.175002>
11. R Abimaheshwari, **R Abinaya**, J Archana, M Navaneethan, S Harish "**Controlled growth of edge site enriched vertical SnS₂ nanoflakes for highly sensitive room temperature detection of NO₂ sensor.**" *Applied Surface Science* 661, 159991 (2024). <https://doi.org/10.1016/j.apsusc.2024.159991>
12. C Archana, **A Rengarajan**, J Archana, M Navaneethan, S Harish "**Realization of low potential barrier in MoS₂/rGO heterojunction with enhanced electrical conductivity for thin film thermoelectric applications.**" *Nanotechnology* 35 (20), 205403 (2024). <https://doi.org/10.1088/1361-6528/ad2630>
13. **R Abinaya**, E Vinoth, S Harish, S Ponnusamy, J Archana, M Shimomura, M Navaneethan "**Modulating Fermi energy in few-layer MoS₂ via metal passivation with enhanced detectivity for near IR photodetector.**" *Journal of Materials Chemistry C* 12 (14), 5247–5256 (2024). <https://doi.org/10.1039/D3TC04603G>

2023

14. SS John, **A Rengarajan**, SK Eswaran, A Jayaram, N Mani "**Engineering phonon transport in MoS₂ lattice by substituting isovalent anion (Se) for low thermal conductivity via all-scale hierarchical phonon scattering.**" *Emergent Materials* 6 (6), 1751–1762 (2023). <https://doi.org/10.1007/s42247-023-00561-3>
15. R Santhosh, **R Abinaya**, S Ponnusamy, H Ikeda, M Navaneethan "Interface scattering induced low thermal conductivity in Ag₂Se/MWCNT for enhanced thermoelectric application." *Diamond and Related Materials* 140, 110344 (2023). <https://doi.org/10.1016/j.diamond.2023.110344>
16. MJK Peer, **A Rengarajan**, N Mani, H Santhana Krishnan "Elucidation of superior thermoelectric transport properties via tuning hopping potential barrier, mass, and strain field fluctuations in LaCo_{0.95}A_{0.05}O₃ (A = Y, In)." *The Journal of Physical Chemistry C* 127 (24), 11707–11711 (2023). <https://doi.org/10.1021/acs.jpcc.3c02130>
17. SS John, **R Abinaya**, S Harish, SK Eswaran, J Archana, M Navaneethan "Influence of secondary phases of Sb₂O₃ and CoSb₂ on CoSb₃ skutterudite to reduce thermal conductivity." *Journal of Alloys and Compounds* 936, 167801 (2023). <https://doi.org/10.1016/j.jallcom.2022.167801>
18. R Santhosh, S Harish, **R Abinaya**, S Ponnusamy, H Ikeda, J Archana, M Navaneethan "Enhanced thermoelectric performance of hot-pressed n-type Ag₂Se nanostructures by controlling the intrinsic lattice defects." *CrystrEngComm* 25 (22), 3317–3327 (2023). <https://doi.org/10.1039/D3CE00195D>

2022

19. C Archana, S Harish, **R Abinaya**, J Archana, M Navaneethan "**Interface modified MoS₂/CNT with enhanced power factor via energy filtering effect for flexible thermoelectric applications.**" *Sensors and Actuators A: Physical* 348, 113938 (2022). <https://doi.org/10.1016/j.sna.2022.113938>

20. **R Abinaya**, S Harish, J Archana, M Shimomura, M Navaneethan "Decoupling trade-off thermoelectric relations and controlled out-plane lattice dynamics in few-layer MoS₂." Applied Physics Letters 121 (12), 123102 (2022). <https://doi.org/10.1063/5.0098848>
21. K Monikapani, V Vijay, **R Abinaya**, J Archana, S Harish, M Navaneethan "Realizing an enhanced Seebeck coefficient and extremely low thermal conductivity in anharmonic Sb-substituted SnSe nanostructures." Journal of Alloys and Compounds 923, 165961 (2022). <https://doi.org/10.1016/j.jallcom.2022.165961>
22. R Santhosh, **R Abinaya**, J Archana, S Ponnusamy, S Harish, M Navaneethan "Controlled grain boundary interfaces of reduced graphene oxide in Ag₂Se matrix for low lattice thermal conductivity and enhanced power factor for thermoelectric applications." Journal of Power Sources 525, 231045 (2022). <https://doi.org/10.1016/j.jpowsour.2022.231045>

2021

23. **R Abinaya**, S Harish, S Ponnusamy, M Shimomura, M Navaneethan, J Archana "Enhanced thermoelectric figure-of-merit of MoS₂/α-MoO₃ nanosheets via tuning of sulphur vacancies." Chemical Engineering Journal 416, 128484 (2021). <https://doi.org/10.1016/j.cej.2021.128484>
24. **R Abinaya**, J Archana, S Harish, M Navaneethan, C Muthamizhchelvan, S Ponnusamy, H Udono, R Sugahara, Y Hayakawa, M Shimomura "Interface driven energy-filtering and phonon scattering of polyaniline incorporated ultrathin layered molybdenum disulphide nanosheets for promising thermoelectric performance." Journal of Colloid and Interface Science 584, 295–309 (2021). <https://doi.org/10.1016/j.jcis.2020.09.096>

2020

25. S Sugarthi, G Bakiyaraj, **R Abinaya**, M Navaneethan, J Archana, M Shimomura "Effect of different growth temperature on the formation of ternary metal chalcogenides AgBiS₂." Materials Science in Semiconductor Processing 107, 104781 (2020). <https://doi.org/10.1016/j.mssp.2019.104781>
26. D Joseph, M Navaneethan, **R Abinaya**, S Harish, J Archana, S Ponnusamy, K Hara, Y Hayakawa "Thermoelectric performance of Cu-doped MoS₂ layered nanosheets for low grade waste heat recovery." Applied Surface Science 505, 144066 (2020). <https://doi.org/10.1016/j.apsusc.2019.144066>

2018

27. S Harish, AN Naveen, **R Abinaya**, J Archana, R Ramesh, M Navaneethan, M Shimomura, Y Hayakawa "Enhanced performance on capacity retention of hierarchical NiS hexagonal nanoplate for highly stable asymmetric supercapacitor." Electrochimica Acta 283, 1053–1062 (2018). <https://doi.org/10.1016/j.electacta.2018.07.017>
28. **R Abinaya**, J Archana, S Harish, M Navaneethan, S Ponnusamy, C Muthamizhchelvan, M Shimomura, Y Hayakawa "Ultrathin layered MoS₂ nanosheets with rich active sites for enhanced visible light photocatalytic activity." RSC Advances 8 (47), 26664–26675 (2018). <https://doi.org/10.1039/C8RA04627A>

2017

29. M Navaneethan, S Nithiananth, **R Abinaya**, S Harish, J Archana, L Sudha, S Ponnusamy, C Muthamizhchelvan, H Ikeda, Y Hayakawa "Hydrothermal growth of highly monodispersed TiO₂ nanoparticles: functional properties and dye-sensitized solar cell performance." Applied Surface Science 418, 186–193 (2017). <https://doi.org/10.1016/j.apsusc.2016.11.167>

B. MANUSCRIPTS UNDER PREPARATION

1. **R. Abinaya**‡, G. Paulraj, S. Harish, M. Navaneethan, J. Archana, M. Shimomura, "Controlled Growth of Horizontal and Honeycomb-Like Vertically Aligned MoS₂ Nanosheets for Efficient Thermoelectric Performance", To be submitted to ACS Nano.
2. **R. Abinaya**‡, Sagarika Sharma, Oleksandr Cherniushok, Taras Parashchuk, and Krzysztof Wojciechowski‡, "Interface Engineering in p-Type Argyrodite Cu_{7.95}SiS₃Se₃ Thermoelectric Legs: Electrode Compatibility, Diffusion Behavior, and Seebeck Coefficient Mapping", To be submitted to The Journal of Physical Chemistry Letters.
3. **R. Abinaya**‡, R. Santhosh, Simon Sajan John, J. Archana‡, M. Navaneethan‡, "Designing double barrier and low-density interfaces in MoS₂ embedded rGO/PEDOT: PSS ternary nanocomposites for thermoelectric applications", To be submitted to ACS Applied Energy Materials.

PUBLICATION COVERAGE BY REQUIRED RESEARCH DOMAIN

The table below maps publications directly to each domain required by the position.

Domain	Key Publications
Thermoelectrics (2D materials)	APL (2022), Chem. Eng. J. (2021), J. Colloids Interface Sci. (2020), J. Mater. Chem. C (2024), ACS Appl. Energy Mater. (2025), Emergent Mater. (2023), CrystEngComm (2023), J. Power Sources (2022), Sensors & Actuators A (2022)
Solar Cells / Photocatalysis	TiO ₂ dye-sensitized solar cell: Appl. Surface Sci. (2017); Graphene–Si Schottky solar cell (M.Sc. thesis); MoS ₂ photocatalysis: RSC Advances (2018)
Photodetectors	Near-IR photodetector (MoS ₂): J. Mater. Chem. C 12, 5247 (2024)
Gas Sensors	SnS ₂ /SnS NO ₂ sensor: J. Alloys Compd. 1001, 175002 (2024); SnS ₂ NO ₂ : Appl. Surface Sci. 661, 159991 (2024)
Electrocatalysis / Energy Storage	NiS supercapacitor: Electrochimica Acta 283, 1053 (2018); Ag ₂ Se/rGO energy device: J. Power Sources 525, 231045 (2022)
2D Materials	All MoS ₂ , WS ₂ , SnS ₂ , MoSe ₂ -based papers (>15 publications); AGH flexible nanocomposite (2026)

SELECTIVE CONFERENCE PRESENTATIONS

Highly relevant to the requirement

Oral Presentations

- 01. Abinaya Rengarajan** "Enhanced charge transfer at zero-barrier injection of MoS₂/α-MoO₃ nanocomposites." 40th Intl. / 20th European Thermoelectric Conference, Kraków, Poland (Jun–Jul 2024).
- 02. Abinaya Rengarajan** "Enhancing thermoelectric properties by coherent interface of MoS₂/WS₂ nanohybrid." ICONN 2019, SRM IST, India. ★ **Best Oral Presentation Award (1st Place)**
- 03. Abinaya Rengarajan** "Thermoelectric performance of MoS₂ embedded on rGO/PEDOT:PSS nanocomposites." NCHSM-2019, SRM IST, India. ★ **Best Oral Presentation Award (3rd Place)**
- 04. Abinaya Rengarajan** "Vertically aligned MoS₂ thin film for thermoelectric application." ICNERE 2018, Shizuoka University, Japan. ★ **Young Researcher Award & Best Oral Presentation**

Poster Presentations

- P1. Abinaya Rengarajan**, et al. "Controlled carrier-phonon dynamics in MoS₂/WO₃ nanohybrid." ICONN 2024, SRM IST, Chennai, India.
- P2. Abinaya Rengarajan**, et al. "Thermoelectric transports in chemical-free fabricated non-periodic MoS₂ holey structure." ICONN 2024, SRM IST, Chennai, India.
- P3. Abinaya Rengarajan**, et al. "Highly enhanced thermoelectric power factor of MoS₂ by MoO₂ metallic inclusion." ICONN 2019, SRM IST, India. ★ **Best Poster Award (1st Place)**
- P4. Abinaya Rengarajan**, et al. "Facile synthesis of layered MoS₂ nanostructures for thermoelectric application." National Symposium, Periyar University, India (2017). ★ **Best Poster Award (1st Place)**
- P5. Abinaya Rengarajan**, et al. "Graphene Schottky junction solar cell." 2nd National Conference on Nanophotonics, Bharathidasan University, India (2016).

RESEARCH EXPERTISE AND INSTRUMENTATION

Materials Synthesis

- CVD (2-zone, 3-zone) growth of 2D TMDCs (MoS₂, WS₂, MoS₂-MoO_x, SnS₂, 2D heterostructures) — monolayer to bulk, lateral and vertical heterostructures
- Hydrothermal synthesis, solution-processed nanocomposites (inorganic–inorganic and inorganic–organic)
- Arc melting, planetary ball milling, spark-plasma sintering (SPS), hot press, spray pyrolysis, spin coating, doctor blade, thermal/e-beam evaporation, in-situ polymerization

Characterization Techniques and Data Analysis

- Thermoelectric transport: Seebeck & resistivity (ZEM-3, home-built high-T system), LFA 467HT (thermal diffusivity), Van der Pauw Hall (HL5500 PC), I–V (Keithley 2400), quantum efficiency
- Structural: XRD (RINT-Ultima III), FESEM (JEOL JSM-7001F), STEM/TEM (JEOL JEM-2100F), EPMA (JEOL JXA-8530F), AFM (SPI3800 non-contact)

- Spectroscopic: Raman (LAB-RAM NRS-7100), XPS (ESCA / Shimadzu Axis Ultra DLD), UV-Vis (JASCO V-750), TGA (DTG-60A)
- Self-built: High-temperature (300–750 K) Seebeck coefficient (2-probe) and linear 4-probe electrical resistivity system; calibrated to >99 % with constantan standard
- Self-built: set-up for Ion migration realization in solid-state materials towards high-performance thermoelectric devices
- Software used: Origin, MS office, XPS peak41, GATAN, Image J, XEI-SW, XpertHighScore,

STUDENT SUPERVISION

Co-supervised the following Ph.D. students at SRM Institute of Science and Technology, India (2021–2024):

- Mr. Santhosh R — Ph.D., SRM IST (publications in J. Power Sources, Diamond & Related Mater., CrystEngComm)
- Ms. Archana C — Ph.D., SRM IST (publications in Sensors & Actuators A, ACS Appl. Energy Mater., Carbon Letters)
- Mr. Simon Sajan John R — Ph.D., SRM IST (publications in Emergent Mater., J. Alloys Compd.)
- Ms. Abimaheswari R — Ph.D., SRM IST (publications in J. Alloys Compd., Appl. Surface Sci.)

PROFESSIONAL SERVICE AND MEMBERSHIPS

- Organization Team Member, Summer Thermoelectric School (2024), Kraków, Poland
- Organization Team Member of International Thermoelectric Conference (2024), Kraków, Poland
- Organization Team Member, International Conference on Advanced Functional Materials and Devices (2024), Chennai, India
- Member, Japan Society of Applied Physics (JSAP), 2020 – present
- President, Tamil Nadu Youth Science Forum (2015–2016), Tiruchirappalli Zone, India
- Member, Tamil Nadu Science Forum (TNSF) | Member, International Awards for Young People (IAYP),
- Volunteer, Science Promotion in Schools across Tamil Nadu (2014 – 2017), India
- Active Reviewer in journals from ACS, IOP, Elsevier, Springer Nature

WORKSHOPS AND COURSES

- Active participant of team/department seminars weekly/monthly under international environment (Poland, India, Japan)
- Delivered seminar at SRM IST (2019 & 2021), India
- Delivered seminar at AGH WiMiC department (Nov 2024), Poland
- Delivered Seminar “Application of artificial intelligence in simulation and control of Metallurgical processes: NeuroMet2025”, Poland
- Completed professional development: Nature Masterclass courses on research paper writing & review writing (2024)
- On-demand web-seminar “Introduction to Thermoelectric conversion” (February,1 – March,14), 2021
- Workshop on crystallographic calculations with WinCSD (2024). by Prof. Yuri Grin, Max Planck Institute
- **Acquired Vast knowledge by active participation in webinar for grants: MSCA NCP, NCN, ERC, NAWA**
- Coursera: Python, Matlab

LANGUAGES

- **Tamil** (Native)
- **English** (fluent)
- **Hindi** (writing & conversational)
- **Japanese** (writing & moderate)
- **Polish** (basics)

Funding Acquisition

Supporting supervisor by leading team for funding acquisition, Total Projects: 11

S.No	Project Title	PI / Co-PI	Period	Funding Agency	Amount (Euros)	Status
1	High-performance thermoelectric materials and devices based on layered MoS ₂ nanostructures for mid-temperature energy harvesting applications	Dr. J. Archana (PI) Dr. M. Navaneethan (Co-PI) Project Implementation Group	2023–2028	DST FIST	~ 150449	Ongoing
2	Development of high-performance phonon-liquid electron crystal argyrodite compounds via band and entropy engineering for mid-temperature thermoelectric generator	Dr. M. Navaneethan (PI)	2023–2026	SERB (CRG)	~ 34915	Ongoing
3	Design and development of Silver-based wearable thermoelectric generator for self-powered health monitoring devices	Dr. S. Harish (PI) Dr. M. Navaneethan (Co-PI)	2023–2026	DST-SERB (CRG)	~ 32828	Ongoing
4	Development of high-performance silicide-based thermoelectric materials and devices for mid-temperature applications	Dr. J. Archana (PI) Dr. M. Navaneethan (Co-PI)	2022–2025	SERB (CRG)	~ 30324	Completed
5	Designing of two-dimensional MoS ₂ –Bi ₂ Se ₃ layered nanocomposites with nano-inclusions for Thermoelectric Applications	Dr. M. Navaneethan (PI)	2023–2026	CSIR EMR	~ 27026	Ongoing
6	Growth of tin-selenide (SnSe) polycrystal to fabricate high-efficient thermoelectric power generator for green energy production	Dr. S. Harish (PI)	2023–2024	SRM-SERI	~ 4504	Ongoing
7	Development of Nanostructured Materials for Fabrication of Wearable Thermoelectric Generator	Dr. M. Navaneethan (PI)	2022–2025	UGC-DAE CSR	~ 8108	Ongoing

Note: Ongoing = project currently active; Completed = project concluded. Funding agencies: DST = Dept. of Science & Technology; SERB = Science & Engineering Research Board; CSIR = Council of Scientific & Industrial Research; DAE BRNS = Dept. of Atomic Energy, Board of Research in Nuclear Sciences; UGC-DAE CSR = University Grants Commission; SRM-SERI = SRM Internal Research Grant.

REFERENCES

Available upon request. Key referees:

Dr. J. Archana Professor & Head (Research), Ph.D. Supervisor CeMAT, Dept. of Physics & Nanotechnology, SRM IST, Chennai jayarama.archana@gmail.com +91 9659284378	Prof. Masaru Shimomura Professor, Ph.D. Supervisor Grad. School of Sci. & Tech., Shizuoka University, Japan shimomura.masaru@shizuoka.ac.jp +81-53-478-1334
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Dr. M. Navaneethan Professor & Head, Nanotechnology Research Centre (Collaborator) SRM Institute of Science and Technology, Chennai m.navaneethan@gmail.com +91-8754420369	Prof. S. Ponnusamy Registrar & Professor, Post-doc Supervisor Dept. of Physics & Nanotechnology, SRM IST, Chennai suruponnus@gmail.com
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